

Transformer-Based Classification of Cybercrime Complaints in Bangla

Abstract

The rapid growth of digital services has led to a significant increase in cybercrime incidents in Bangladesh. Automatic classification of Bangla cybercrime complaints can support faster complaint management and decision-making. This study evaluates five text classification models—Naive Bayes, BiLSTM, BERT, BanglaBERT, and XLM-R—on a manually annotated Bangla cybercrime complaint dataset. Standard preprocessing techniques, tokenization, and train-validation-test splitting were applied before model training. The models were evaluated using Accuracy, Precision, Recall, and F1-score. BiLSTM achieved the best overall performance with an accuracy of 89%, outperforming the transformer-based and traditional machine learning models. The findings demonstrate that deep contextual learning significantly improves Bangla cybercrime complaint classification.

Keywords

Bangla NLP, Cybercrime Classification, BiLSTM, BanglaBERT, XLM-R, BERT, Naive Bayes, Text Classification.

I. Introduction

The rapid expansion of internet connectivity, smartphone usage, and digital financial services has transformed the way people communicate and conduct daily activities. While these technological advancements have provided significant benefits, they have also created new opportunities for cybercriminals. Cybercrimes such as phishing, online fraud, identity theft, account hacking, cyber harassment, fake profiles, malware attacks, and financial scams have become increasingly common. In Bangladesh, the growing number of online users has resulted in a significant increase in cybercrime complaints submitted through various digital platforms. As a result, effective methods for analyzing and categorizing these complaints have become an important research area.

Traditionally, cybercrime complaints are reviewed and categorized manually by investigators or support personnel. However, manual processing is time-consuming, labor-intensive, and often inconsistent, especially when handling a large volume of complaints. Delays in identifying the type of cybercrime may slow down investigations and reduce the efficiency of law enforcement agencies. Therefore, developing an automated text classification system capable of accurately categorizing Bangla cybercrime complaints can significantly improve the complaint management process and support faster decision-making.

Recent advances in Natural Language Processing (NLP) and deep learning have revolutionized text classification tasks. Traditional machine learning algorithms such as Naive Bayes have been widely used because of their simplicity, computational efficiency, and ability to establish strong baseline performance. However, these methods rely heavily on manually engineered features and often fail to capture complex contextual relationships within natural language. As a result, their performance is limited when dealing with semantically rich Bangla text.

Deep learning models, particularly Bidirectional Long Short-Term Memory (BiLSTM), overcome many of these limitations by learning contextual information from both forward and backward directions of a sentence. BiLSTM effectively captures sequential dependencies and long-range relationships between words, making it highly suitable for text classification tasks. More recently, transformer-based language models such as BERT, BanglaBERT, and XLM-RoBERTa have achieved remarkable success across various NLP applications by learning contextual word representations through self-attention mechanisms. BanglaBERT is specifically pre-trained on large-scale Bangla corpora, enabling it to better understand the linguistic characteristics of the Bangla language, while XLM-RoBERTa provides multilingual language understanding across numerous languages.

Although several studies have explored Bangla text classification using machine learning and transformer-based approaches, limited research has specifically focused on Bangla cybercrime complaint classification using a comprehensive comparison of traditional machine learning, deep learning, and transformer models. Furthermore, the increasing diversity of cybercrime categories requires models capable of accurately distinguishing complaints with similar linguistic patterns. Therefore, a comparative evaluation of different classification techniques is essential to determine the most effective approach for this application.

In this study, a comparative analysis of five classification models—Naive Bayes, BiLSTM, BERT, BanglaBERT, and XLM-RoBERTa—is conducted on a manually annotated Bangla cybercrime complaint dataset. The dataset is preprocessed through text cleaning and tokenization before being divided into training, validation, and testing sets. The performance of each model is evaluated using Accuracy, Precision, Recall, and F1-score to ensure a fair comparison.

Experimental results demonstrate that the BiLSTM model achieved the highest classification accuracy of **89%**, outperforming the transformer-based models BanglaBERT (**85.78%**) and XLM-R (**85.55%**), as well as Naive Bayes (**75%**) and BERT (**67%**). These findings indicate that BiLSTM effectively captures contextual and sequential information present in Bangla cybercrime complaints and provides the most suitable solution for this dataset. The outcomes of this research may contribute to the development of intelligent cybercrime complaint management systems that assist law enforcement agencies and improve the efficiency of cybercrime investigation in Bangladesh.

II. Related Work

Natural Language Processing (NLP) has become an essential technology for automatically analyzing and classifying textual data. In recent years, researchers have applied machine learning, deep learning, and transformer-based language models to various Bangla text

classification tasks, including sentiment analysis, fake news detection, hate speech identification, cyberbullying detection, and cybercrime-related text analysis.

In [1], the authors proposed a machine learning-based approach for Bangla text classification using traditional algorithms such as Naive Bayes, Support Vector Machine (SVM), and Random Forest. Their study demonstrated that conventional machine learning methods provide efficient baseline performance with relatively low computational cost. However, the models struggled to capture contextual and semantic relationships within Bangla text, resulting in lower classification accuracy compared to deep learning approaches.

Deep learning techniques have significantly improved the performance of Bangla NLP applications. In [2], the authors employed Long Short-Term Memory (LSTM) and Bidirectional Long Short-Term Memory (BiLSTM) networks for Bangla text classification. Their results showed that BiLSTM outperformed conventional machine learning models by learning contextual information from both forward and backward directions of a sentence. The study concluded that sequential neural architectures are more effective for capturing linguistic patterns in Bangla.

Transformer-based language models have recently become the state-of-the-art approach for many NLP tasks. In [3], BERT was fine-tuned for Bangla text classification and achieved superior contextual understanding compared to recurrent neural networks.

Although these studies have reported promising results for Bangla NLP, relatively few have focused specifically on Bangla cybercrime complaint classification while providing a comprehensive comparison of traditional machine learning, deep learning, and transformer-based models under the same experimental setting. To address this research gap, the present work evaluates Naive Bayes, BiLSTM, BERT, BanglaBERT, and XLM-R using the same dataset, preprocessing pipeline, and evaluation metrics. This comparative analysis provides a clearer understanding of the strengths and limitations of each model for Bangla cybercrime complaint classification.

III. Dataset Description

The dataset used in this study consists of **1,500 manually annotated Bangla cybercrime complaints** collected manually. The complaint categories were carefully designed according to the legal definitions and guidelines provided in the Bangladesh Information and Communication Technology (ICT) Act, 2006 and the Right to Information Act, 2009. The dataset was developed to support automatic classification of cybercrime-related incidents reported in the Bangla language. Each complaint describes a specific cybercrime incident and is assigned to a predefined category based on its primary context.

To ensure annotation quality and consistency, the complaints were manually reviewed and labeled according to the definitions of different cybercrime categories. The annotation process followed relevant cybersecurity concepts and legal guidelines to maintain reliable class assignments. Each sample contains two primary fields: the complaint text and its corresponding class label.

The dataset includes complaints from multiple cybercrime categories, such as **Account Hacking, Phishing, Online Scam, OTP Fraud, Financial Fraud, Fake Profile, Identity Theft, Image Misuse, Cyber Harassment, Cyber Stalking, Cyber Blackmail, Malware Attack, Data Breach, Copyright Violation, Romance Scam, and Cyber Abuse**. These categories represent common cybercrime incidents frequently reported in Bangladesh.

Before model training, several preprocessing techniques were applied to improve data quality. The preprocessing pipeline included text normalization, removal of unnecessary symbols and punctuation, elimination of duplicate records, handling of missing values, and tokenization. The cleaned dataset was then divided into **training, validation, and testing sets** to ensure an unbiased evaluation of all classification models.

The prepared dataset was used to train and evaluate five different classification models: **Naive Bayes, BiLSTM, BERT, BanglaBERT, and XLM-R**. All models were trained and tested using the same dataset split and preprocessing procedure to ensure a fair performance comparison. The effectiveness of each model was assessed using four standard evaluation metrics: **Accuracy, Precision, Recall, and F1-score**.

IV. Methodology

This study follows a systematic methodology for the automatic classification of Bangla cybercrime complaints. The overall workflow consists of dataset collection, data preprocessing, model development, model training, and performance evaluation. A comparative analysis was conducted using five different classification models: Naive Bayes, BiLSTM, BERT, BanglaBERT, and XLM-R.

A. Data Preprocessing

The collected dataset was preprocessed to improve data quality and ensure compatibility with the classification models. Initially, duplicate and incomplete records were identified and removed. The complaint texts were then cleaned by eliminating unnecessary punctuation marks, special symbols, extra white spaces, URLs, and non-textual characters that did not contribute to the semantic meaning of the complaints.

After cleaning, Bangla text normalization was performed to standardize character representations and reduce inconsistencies caused by different writing styles. The normalized text was then tokenized using the tokenizer appropriate for each model. For the transformer-based models (BERT, BanglaBERT, and XLM-R), the corresponding pre-trained tokenizers were used to convert the complaint text into token IDs and attention masks. For the BiLSTM model, the text was converted into integer sequences using a vocabulary generated from the training data, followed by sequence padding to obtain a fixed input length. The Naive Bayes model utilized vectorized text representations generated from the preprocessed corpus.

Finally, the dataset was divided into training, validation, and testing subsets. The training set was used to learn model parameters, the validation set was used for hyperparameter tuning and monitoring model performance, and the testing set was reserved for the final evaluation.

B. Model Development

Five classification models representing different learning paradigms were implemented and compared.

Naive Bayes was selected as the traditional machine learning baseline because of its computational efficiency and simplicity in text classification.

BiLSTM (Bidirectional Long Short-Term Memory) was employed to capture contextual dependencies by processing each sentence in both forward and backward directions. This architecture enables the model to learn long-range sequential information from Bangla cybercrime complaints.

BERT (Bidirectional Encoder Representations from Transformers) was fine-tuned for the classification task using its pre-trained contextual language representations. The model utilizes self-attention mechanisms to capture semantic relationships between words.

BanglaBERT, a transformer model pre-trained specifically on large-scale Bangla corpora, was included to evaluate the effectiveness of language-specific pre-training for Bangla cybercrime complaint classification.

XLM-R (XLM-RoBERTa), a multilingual transformer model trained on more than one hundred languages, was incorporated to investigate whether multilingual contextual representations can effectively classify Bangla cybercrime complaints.

All models were trained using the same training dataset and evaluated under identical experimental conditions to ensure a fair comparison.

C. Performance Evaluation

The performance of each classification model was evaluated using four widely adopted metrics: **Accuracy**, **Precision**, **Recall**, and **F1-score**.

Accuracy measures the proportion of correctly classified complaints among all samples.

Precision evaluates how many of the complaints predicted as a specific cybercrime category were actually correct.

Recall measures the model's ability to identify all complaints belonging to a particular class.

F1-score is the harmonic mean of Precision and Recall and provides a balanced evaluation, particularly when class distributions are uneven.

The experimental results obtained from these evaluation metrics were compared to identify the most suitable model for Bangla cybercrime complaint classification.

V. Results and Discussion

The performance of the five classification models was evaluated using **Accuracy**, **Precision**, **Recall**, and **F1-score**. Table I presents the comparative performance of all models.

Table I. Performance Comparison of the Proposed Models

Model	Accuracy	Precision	Recall	F1-score
BiLSTM	89.00%	84.2%	80.0%	81.0%
BanglaBERT	85.78%	78.0%	73.0%	80.0%
XLM-R	85.55%	79.0%	71.0%	81.0%
Naive Bayes	75.00%	71.0%	69.0%	70.0%
BERT	67.00%	65.0%	69.1%	66.0%

The experimental results show that **BiLSTM** achieved the highest performance with an **accuracy of 89%**, outperforming all other models. **BanglaBERT (85.78%)** and **XLM-R (85.55%)** also produced competitive results, demonstrating the effectiveness of transformer-based models for Bangla text classification. In contrast, **Naive Bayes** and **BERT** achieved lower performance, with accuracies of **75%** and **67%**, respectively.

Overall, the results indicate that **BiLSTM** is the most effective model for the proposed Bangla cybercrime complaint classification task, providing the best balance between accuracy, precision, recall, and F1-score.

VI. Conclusion

This study presented a comparative analysis of five text classification models—Naive Bayes, BiLSTM, BERT, BanglaBERT, and XLM-R—for Bangla cybercrime complaint classification. Experimental results showed that **BiLSTM** achieved the best performance with an **accuracy of 89%**, outperforming the other evaluated models. BanglaBERT and XLM-R also demonstrated strong classification capability, while Naive Bayes and BERT achieved comparatively lower performance.

The findings suggest that BiLSTM is a reliable and effective approach for Bangla cybercrime complaint classification. In future work, the study can be extended by using larger datasets, exploring ensemble and lightweight transformer models, and deploying the system as a real-time cybercrime complaint classification application.

References

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